

DEEPFAKE VIDEO DETECTION — SPATIAL-TEMPORAL CNN/TRANSFORMER WITH CROSS-GENERATOR ARTIFACT ANALYSIS

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ABSTRACT

Deepfake videos—face swaps, re-enactments, and attribute edits produced by generative models—are increasingly used for misinformation and impersonation, yet most detectors are evaluated only on the generators they were trained on. This report presents a deepfake video detection system whose objective is twofold: to classify manipulated videos reliably, and to study *which manipulation artifacts generalize across generators*. A two-stage architecture was developed: an EfficientNet-B0 spatial detector scores individual face frames, and frame-level evidence is aggregated into a video-level verdict either by mean pooling or by a three-layer temporal transformer. The system was trained on FaceForensics++ (five manipulation generators) and evaluated with Celeb-DF v2 as a held-out cross-dataset test, under strictly video-level disjoint splits, a frame-count-controlled protocol, and CPU-only compute. The mean-pool spatial detector achieves a video-level AUC of 0.9553 (91.7% accuracy) on the held-out test split and 0.9033 AUC across sources. Spectral (FFT) and Grad-CAM analyses identify high-frequency upsampling fingerprints as the strongest cue, while a cross-preprocessing study (AUC dropping to 0.6419) shows these artifacts are pipeline-dependent. A privacy-preserving browser demo runs the model entirely on-device via ONNX/WebAssembly.

Index Terms—Deepfake Detection, Digital Forensics, EfficientNet, Temporal Transformer, Cross-Generator Generalization, Spectral Analysis.

1. INTRODUCTION

Generative face-manipulation models can now produce videos that are visually indistinguishable from genuine footage. Such “deepfakes” have been deployed in political

disinformation, financial fraud, and non-consensual impersonation, making automatic detection a problem of practical and societal importance. A large body of forensic methods exists, but most detectors are trained and evaluated on a single manipulation dataset; their accuracy typically collapses when confronted with a generator they have not seen. The core difficulty is that a detector may learn *dataset-specific* artifacts—compression, cropping pipelines, or generator fingerprints—rather than manipulation cues that transfer.

This project develops a deepfake video detection system and, beyond accuracy, asks a research question: *which manipulation artifacts generalize across generators?* The system combines an EfficientNet-B0 spatial detector that scores individual face frames with a temporal transformer that aggregates frame embeddings into a video-level verdict. FFT spectral and Grad-CAM attention analyses characterize the cues the detector relies on, and a deliberately hard cross-preprocessing evaluation quantifies how those cues degrade when the face-crop pipeline differs from training.

All experiments use CPU-only hardware, and the model is deployed as a fully on-device browser demo. Section 2 defines the problem and objectives, Section 3 details the methodology, Section 4 presents results, Section 5 discusses them, and Sections 6–7 conclude and list reproducible artifacts.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Given an input video, the task is binary video-level classification: decide whether the footage is genuine or has been manipulated by a face-synthesis method (face swap, re-enactment,

attribute transfer, or entire-face generation). The scope is forensic screening of facial manipulation in compressed video; audio forensics and full-frame scene manipulation are out of scope. Three aspects make the problem hard: public dataset mirrors contain split leakage and frame-count shortcuts that silently inflate accuracy, so a trustworthy evaluation protocol must precede any model comparison; manipulation artifacts differ by generator family (swap-based methods modify only the face interior, autoencoder-based methods resynthesize it); and any deployed detector must run without uploading private footage.

2.2. Objectives

The specific objectives were: (1) to study existing deepfake detection approaches and datasets; (2) to design and implement a spatial-temporal detector—an EfficientNet-B0 frame scorer with temporal aggregation—trainable on CPU; (3) to construct leak-free, video-level disjoint train/val/test splits and a length-control protocol that removes the frame-count shortcut; (4) to evaluate video-level performance with AUC, average precision, and accuracy, including cross-dataset generalization to Celeb-DF v2; (5) to analyze which artifacts generalize using FFT spectral divergence and Grad-CAM attention; and (6) to deploy the detector as a zero-backend, on-device browser application.

3. METHODOLOGY / APPROACH

3.1. Overall Workflow

Figure 1 summarizes the pipeline. From each video, 16 frames are evenly sampled; a Haar-cascade face detector with a bounding-box margin extracts face crops. The spatial branch passes every crop through EfficientNet-B0 (ImageNet-initialized) to produce a per-frame fake probability. For video-level inference, frame scores are aggregated by mean pooling. As an alternative, frame embeddings (dimension 1280) are cached and consumed by a temporal transformer that learns to aggregate over the frame sequence.

3.2. Technical Details

Spatial detector. EfficientNet-B0 [3] was chosen for its accuracy-to-size ratio under CPU budgets. Training proceeds in two phases to keep compute tractable: first the classification head only (binary cross-entropy loss), then full fine-tuning on a class-balanced subset. Validation AUC improved from 0.7910 (head-only) to 0.8750 after fine-tuning and to 0.9100 after balanced fine-tuning.

Temporal aggregation. Frame embeddings are projected to 256 dimensions, augmented with positional embeddings, and processed by a 3-layer, 4-head transformer [4] with a CLS token and attention pooling, producing a video-level fake probability. Because clip lengths differ systematically between classes in the data mirror (Section 5), all temporal experiments use a length-control protocol: every clip is subsampled to exactly $T = 2$ frames and shorter clips are dropped, so the model cannot exploit clip length as a feature.

Artifact analysis. Two probes answer the generalization question. (1) *Spectral fingerprints:* for each source, the radial power spectrum of fake frames is compared against real frames via an L1 divergence, decomposed into low/mid/high frequency bands; GAN-style decoders are known to leave periodic upsampling structure in the mid/high bands [6], [7]. (2) *Spatial attention:* Grad-CAM [5] maps are reduced to a center-of-mass statistic (0 = image center) to test whether the detector attends to the manipulated face interior or to borders and compression.

3.3. Project Architecture

The implementation is a single Python package (`deepfake_detection`) with YAML-configured stages: `prepare_data.py` builds leak-free video-disjoint splits; `models/spatial.py` and `models/temporal.py` define the two networks; `train_spatial.py` and `train_temporal.py` run training and AUC evaluation; `analyze_generalization.py` and `analyze_artifacts.py` produce the per-generator study and FFT/Grad-CAM reports; and `export_onnx.py` exports an fp16 ONNX model consumed by the static browser app, which pairs MediaPipe BlazeFace [9] with ONNX Runtime Web (WASM) for fully client-side inference. Unit tests cover parsing, models, metrics, and preprocessing.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

Data. The primary corpus is pre-extracted face frames from FaceForensics++ (C23) [1] covering five generators (Deepfakes, Face2Face, FaceSwap, NeuralTextures, FaceShifter), with Celeb-DF v2 [2] as a held-out, out-of-distribution test source; face crops use a Haar cascade [8] with margin. *Protocol.* Splits are video-level disjoint (1,044 leaked training rows dropped); all models train on CPU. *Metrics.* Video-level AUC (threshold-independent), average precision (AP), and accuracy; the held-out test set comprises 1,289 videos.

4.2. Results

Video-level benchmark. Table 1 reports the main results under length control ($T = 2$); mean-pooling of spatial frame scores is the strongest configuration.

TABLE 1
VIDEO-LEVEL BENCHMARK ON THE HELD-OUT TEST SPLIT ($n = 1289$,
LENGTH-CONTROLLED $T = 2$). BEST RESULT IN BOLD.

Model / aggregation	AUC	AP	Acc.
Spatial, mean-pool	0.9553	0.9838	0.9170
Spatial, max-pool	0.9336	0.9764	0.8960
Spatial, top- k	0.9336	0.9764	0.8960
Spatial + temporal transformer	0.9456	0.9768	0.8813

Cross-source generalization. With one model trained on the FaceForensics++ mix, Table 2 shows performance per held-out source, including 0.9220 AUC on the out-of-distribution Celeb-DF v2 family.

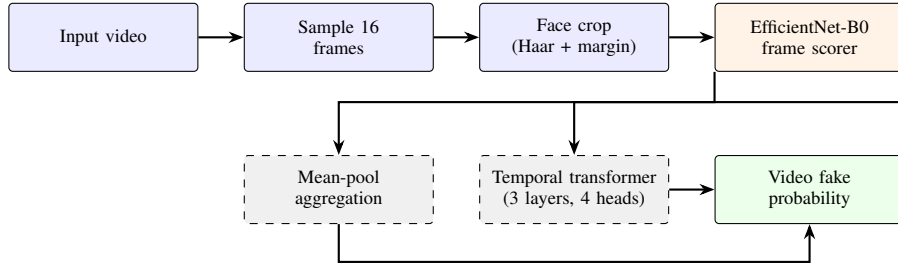


Fig. 1. Overview of the detection pipeline. Sixteen evenly sampled frames are face-cropped and scored independently by EfficientNet-B0. Video-level evidence is aggregated either by mean pooling of frame scores (best result) or by a temporal transformer over frame embeddings; both paths yield a single fake probability for the whole video.

TABLE 2
CROSS-SOURCE GENERALIZATION OF A SINGLE MODEL ON HELD-OUT TEST VIDEOS.

Source	Test AUC	n
Celeb-DF v2 (OOD family)	0.9220	3,000
FaceForensics++	0.8928	1,916
Overall	0.9033	–

Cross-preprocessing generalization. When test frames come from a *different* face-crop pipeline than training, per-generator results (Table 3) degrade sharply: autoencoder-based Deepfakes remain detectable (0.8011 AUC) while swap-based forgeries fall to 0.56–0.63 AUC.

TABLE 3
PER-GENERATOR AUC UNDER CROSS-PREPROCESSING EVALUATION (DIFFERENT FACE-CROP PIPELINE THAN TRAINING).

Generator	Test AUC	n
Deepfakes	0.8011	1,394
Face2Face	0.6291	1,399
NeuralTextures	0.6220	1,399
FaceShifter	0.5955	1,398
FaceSwap	0.5629	1,398
Overall	0.6419	6,988

Artifact analysis. FF++ fakes deviate from real frames $2.3\times$ more in the radial power spectrum than Celeb-DF fakes (L1 0.054 vs. 0.024), driven by the high-frequency band (0.025 vs. 0.009)—the classic decoder upsampling fingerprint. Grad-CAM center-of-mass statistics show fakes pull attention into the face interior (0.32 Celeb-DF, 0.22 FF++) whereas real frames scatter (0.09), confirming the detector keys on manipulated facial content rather than borders or compression.

5. DISCUSSION

Two data-integrity problems were found and fixed, and both materially affect reported accuracy. The Kaggle mirror’s provided splits leaked 124 validation and 78 test videos into training; without rebuilding video-disjoint splits, every metric in Table 1 would be optimistically biased. Additionally, real clips in the mirror contain 12 frames while fakes contain only

2–3, so a temporal model can achieve a perfect AUC merely by *counting frames*; the length-control protocol ($T = 2$) removes this shortcut.

The mean-pool aggregator outperforming the temporal transformer (0.9553 vs. 0.9456 AUC) is interpretable: under length control each clip contributes very few frames, leaving the transformer little sequential structure, while averaging is a robust, variance-reducing estimator of frame-level evidence. The cross-preprocessing study is the main negative finding: artifacts generalize well *within* a preprocessing pipeline (AUC 0.90–0.96) but degrade sharply *across* pipelines (0.56–0.80); swap-based forgeries leave the weakest cue because they modify only the face interior, whereas autoencoder-based methods resynthesize the whole crop and leave stronger decoder fingerprints. On the deployment side, int8 quantization was rejected (it collapses EfficientNet’s squeeze-and-excitation blocks, dropping frame AUC from 0.97 to 0.55); fp16 export halves model size with no measurable loss. The principal limitation is thus pipeline dependence: a production system would need multi-pipeline training data or preprocessing-invariant features, and detection scores should be read as forensic evidence cues, not legal proof.

6. CONCLUSION AND FUTURE WORK

This project delivered a CPU-trained deepfake detector achieving 0.9553 video-level AUC on a leak-free, length-controlled test split (0.9033 across sources, including an out-of-distribution generator family), an artifact study identifying high-frequency spectral fingerprints as the strongest but pipeline-dependent cue, and a fully on-device browser deployment. The key insight is that within-pipeline benchmark accuracy overstates real-world robustness. Future work includes joint multi-pipeline, multi-generator training, preprocessing-invariant representations, and adversarial robustness testing.

7. ARTIFACTS AND DEMONSTRATIONS

All artifacts are publicly accessible. *Code repository*: <https://github.com/saketkumar-18/deepfake-detection> (full source, configs, unit tests, training logs, and analysis scripts). *Live demo*: <https://deepfake-detector-amber.vercel.app> — a zero-backend browser app running the detector fully on-device via fp16 ONNX in WebAssembly. *Video demonstration*: the accompanying video submission is on YouTube.

8. REFERENCES

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